

# Differential Geometry 5

## Theory: Plane cubics and the Cremona involution

### Plane cubics

Assume  $k$  is algebraically closed.

A *plane cubic* is a curve

$$C \subset \mathbb{P}^2$$

defined by a single homogeneous polynomial of degree 3:

$$C = Z(F), \quad F \in k[x, y, z] \text{ homogeneous of degree 3.}$$

An affine cubic in  $\mathbb{A}^2$  given by  $f(x, y)$  has projective closure obtained by homogenization:

$$F(x, y, z) = z^3 f\left(\frac{x}{z}, \frac{y}{z}\right).$$

### Classification by reducibility and singularities

Every plane cubic falls into one of the following types.

**Smooth cubic.** If  $C$  is smooth, then it is irreducible and has genus

$$g(C) = 1.$$

After choosing a base point  $O \in C$ , it carries the structure of an elliptic curve via the chord–tangent law. A line meets  $C$  in 3 points counting multiplicity.

**Singular irreducible cubic.** If  $C$  is irreducible but singular, then it is rational:

$$g(C) = 0.$$

The typical singularities are a node or a cusp, and  $C$  admits a rational parametrization.

**Reducible cubic.** If  $F$  factors, then  $C$  is reducible. The possibilities are:

three lines (with multiplicities),

or

a line plus a conic (with multiplicities).

For example,

$$Z(xyz) = Z(x) \cup Z(y) \cup Z(z), \quad Z(z(xy - z^2)) = Z(z) \cup Z(xy - z^2).$$

**Bezout's theorem for cubics**

If  $C = Z(F)$  and  $D = Z(G)$  are plane curves in  $\mathbb{P}^2$  with no common component, then

$$\sum_{p \in C \cap D} I_p(C, D) = \deg(F) \deg(G).$$

In particular, a line meets a cubic in

$$3$$

points counting multiplicity, and a conic meets a cubic in

$$6$$

points counting multiplicity.

**Weierstrass form**

Assume  $\text{char}(k) \neq 2, 3$ . Every smooth plane cubic is projectively equivalent to a Weierstrass model

$$y^2z = x^3 + axz^2 + bz^3,$$

with discriminant  $\Delta \neq 0$ . In this form, the group law has the explicit inversion

$$(x : y : z) \longmapsto (x : -y : z).$$

**The standard Cremona involution**

Define the rational map

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \quad (x : y : z) \longmapsto (yz : zx : xy).$$

It is undefined exactly at the three base points

$$p_1 = (1 : 0 : 0), \quad p_2 = (0 : 1 : 0), \quad p_3 = (0 : 0 : 1).$$

Moreover, it contracts the coordinate lines:

$$Z(x) \text{ to } p_1, \quad Z(y) \text{ to } p_2, \quad Z(z) \text{ to } p_3.$$

The map is birational, and it is an involution as a rational map:

$$\sigma \circ \sigma = \text{id}.$$

**Resolution by blow-ups**

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of  $\mathbb{P}^2$  at the three base points  $p_1, p_2, p_3$ . Then  $\sigma$  lifts to a morphism

$$\tilde{\sigma} : \widetilde{\mathbb{P}^2} \rightarrow \widetilde{\mathbb{P}^2}.$$

In the Picard group of  $\widetilde{\mathbb{P}^2}$ , if  $H$  denotes the pullback of the class of a line and  $E_1, E_2, E_3$  are the exceptional divisors, then

$$\sigma^*(H) = 2H - E_1 - E_2 - E_3,$$

$$\sigma^*(E_1) = H - E_2 - E_3, \quad \sigma^*(E_2) = H - E_1 - E_3, \quad \sigma^*(E_3) = H - E_1 - E_2.$$

**Effect on degrees of plane curves**

Let  $C \subset \mathbb{P}^2$  be a curve of degree  $d$ . Assume  $C$  passes through the base points with multiplicities

$$m_1, \quad m_2, \quad m_3.$$

Then the strict transform of  $C$  under the Cremona involution has degree

$$d' = 2d - m_1 - m_2 - m_3.$$

For example, a general line has  $d = 1$  and  $m_1 = m_2 = m_3 = 0$ , hence

$$d' = 2,$$

so it maps to a conic through the three base points. A conic through all three base points has  $d = 2$  and  $m_1 = m_2 = m_3 = 1$ , hence

$$d' = 1,$$

so it maps to a line.

**Remark on cubics**

For a cubic  $d = 3$ , if it passes simply through  $p_1, p_2, p_3$ , then

$$d' = 2 \cdot 3 - 1 - 1 - 1 = 3,$$

so it transforms to another cubic. Higher multiplicities at the base points can decrease the degree after transformation.

**What is a cubic**

A *cubic* is an algebraic set defined by a polynomial equation of total degree 3.

**Affine case.** A cubic hypersurface in  $\mathbb{A}^n$  is

$$X = Z(f) \subset \mathbb{A}^n, \quad f \in k[x_1, \dots, x_n], \quad \deg(f) = 3.$$

**Projective case.** A cubic hypersurface in  $\mathbb{P}^n$  is

$$X = Z(F) \subset \mathbb{P}^n, \quad F \in k[x_0, \dots, x_n] \text{ homogeneous}, \quad \deg(F) = 3.$$

**Curve versus surface.** A hypersurface in  $\mathbb{A}^n$  (or  $\mathbb{P}^n$ ) typically has dimension  $n - 1$ . Hence:

$$\begin{aligned} \text{cubic in } \mathbb{A}^2 \text{ or } \mathbb{P}^2 &\text{ is a curve,} \\ \text{cubic in } \mathbb{A}^3 \text{ or } \mathbb{P}^3 &\text{ is a surface.} \end{aligned}$$

**Affine cubic versus projective cubic: are they the same?**

Let  $k$  be a field.

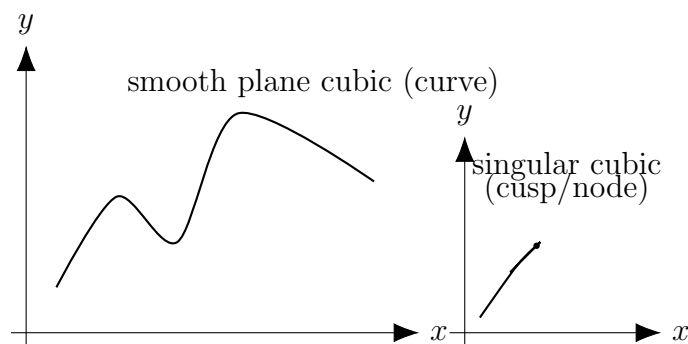


Figure 1: In  $\mathbb{A}^2$  or  $\mathbb{P}^2$ , a cubic is typically a *curve*. Smooth cubics are elliptic curves; singular irreducible cubics are rational.

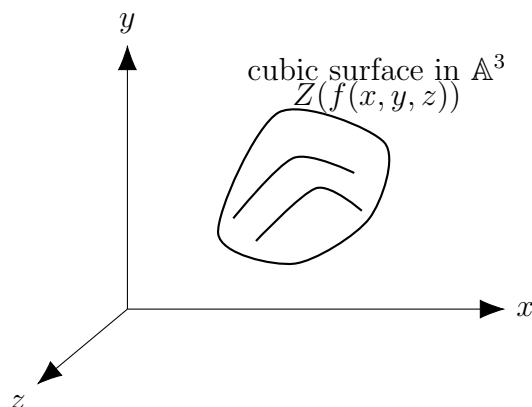


Figure 2: In  $\mathbb{A}^3$  (or  $\mathbb{P}^3$ ), a cubic is typically a *surface*: one degree-3 equation cuts out a 2D object.

### Affine hypersurfaces

A *cubic affine hypersurface* in  $\mathbb{A}^n$  is the zero locus of a polynomial of total degree at most 3:

$$X_{\text{aff}} = Z(f) \subset \mathbb{A}^n, \quad f \in k[x_1, \dots, x_n], \quad \deg(f) \leq 3.$$

### Projective hypersurfaces

A *cubic projective hypersurface* in  $\mathbb{P}^n$  is the zero locus of a homogeneous polynomial of degree 3:

$$X_{\text{proj}} = Z(F) \subset \mathbb{P}^n, \quad F \in k[x_0, x_1, \dots, x_n] \text{ homogeneous}, \quad \deg(F) = 3.$$

In projective space, equations must be homogeneous.

### Homogenization and projective closure

Given  $f \in k[x_1, \dots, x_n]$  with  $\deg(f) \leq 3$ , its homogenization of degree 3 is the homogeneous polynomial

$$F(x_0, x_1, \dots, x_n) = x_0^3 f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Equivalently, each monomial

$$x_1^{a_1} \cdots x_n^{a_n}$$

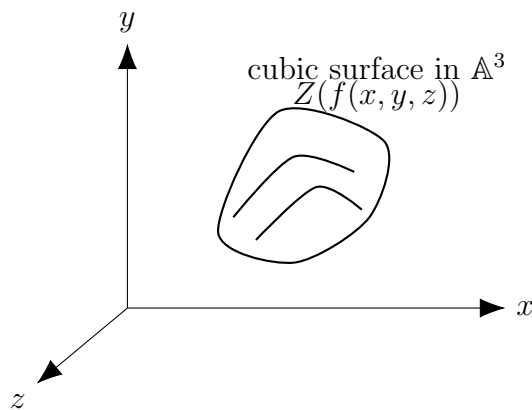


Figure 3: In  $\mathbb{A}^3$  (or  $\mathbb{P}^3$ ), a cubic is typically a *surface*: one degree-3 equation cuts out a 2D object.

of total degree

$$a_1 + \cdots + a_n = d \leq 3$$

is replaced by

$$x_1^{a_1} \cdots x_n^{a_n} x_0^{3-d}.$$

The projective hypersurface

$$X_{\text{proj}} = Z(F) \subset \mathbb{P}^n$$

is called the *projective closure* of

$$X_{\text{aff}} = Z(f) \subset \mathbb{A}^n.$$

### Affine chart and points at infinity

Consider the standard affine chart

$$U_0 = \{x_0 \neq 0\} \subset \mathbb{P}^n,$$

with coordinates

$$x'_i = \frac{x_i}{x_0}.$$

On this chart we have

$$F(1, x_1, \dots, x_n) = f(x_1, \dots, x_n).$$

Hence

$$X_{\text{proj}} \cap U_0 \cong X_{\text{aff}}.$$

So the affine hypersurface is the part of the projective hypersurface lying in the chart  $x_0 \neq 0$ .

However,  $X_{\text{proj}}$  may contain *additional points at infinity*, namely

$$X_{\text{proj}} \cap \{x_0 = 0\}.$$

Thus, in general,

$$X_{\text{proj}} = \overline{X_{\text{aff}}}^{\text{proj}} = X_{\text{aff}} \cup (X_{\text{proj}} \cap \{x_0 = 0\}),$$

where  $\overline{X_{\text{aff}}}^{\text{proj}}$  denotes the projective closure.

**Example**

In  $\mathbb{A}^2$  consider the affine cubic curve

$$f(x, y) = y - x^3.$$

Homogenizing to degree 3 gives a projective cubic in  $\mathbb{P}^2$ :

$$F(x, y, z) = yz^2 - x^3.$$

On the affine chart  $z = 1$ ,

$$F(x, y, 1) = y - x^3,$$

so this recovers the affine curve. At infinity  $z = 0$ ,

$$F(x, y, 0) = -x^3,$$

so  $x = 0$ , and the projective closure contains the point

$$(0 : 1 : 0).$$

Hence the projective cubic is the affine cubic together with an extra point at infinity.

## Why projective closures can be larger although homogeneous polynomials are more restrictive

Let  $k$  be a field.

### Homogeneous polynomials are a smaller class

A polynomial

$$f \in k[x_1, \dots, x_n]$$

is called *homogeneous of degree  $d$*  if every monomial occurring in  $f$  has total degree  $d$ . For example, in  $k[x, y, z]$  the polynomial

$$x^3 + x^2y + yz^2$$

is homogeneous of degree 3, whereas

$$x^3 + y$$

is not homogeneous, because the monomials have degrees 3 and 1.

Thus, as polynomials, homogeneous degree-3 polynomials form a *more restrictive* class than arbitrary degree-3 polynomials.

### Why the geometric set becomes larger in projective space

The apparent paradox comes from comparing objects in *different ambient spaces*. Affine space  $\mathbb{A}^n$  sits inside projective space  $\mathbb{P}^n$  as the standard chart

$$U_0 = \{x_0 \neq 0\} \subset \mathbb{P}^n,$$

via

$$(x_1, \dots, x_n) \mapsto (1 : x_1 : \dots : x_n).$$

Projective space also contains the hyperplane at infinity

$$H_\infty = \{x_0 = 0\} \subset \mathbb{P}^n,$$

which has no affine counterpart.

Given an affine equation

$$f(x_1, \dots, x_n) = 0$$

one forms its homogenization  $F$  and considers the projective hypersurface

$$Z(F) \subset \mathbb{P}^n.$$

On the affine chart  $U_0$  one recovers the original affine hypersurface:

$$Z(F) \cap U_0 \cong Z(f) \subset \mathbb{A}^n.$$

However,  $Z(F)$  may also meet the hyperplane at infinity  $H_\infty$ . Hence, in general,

$$Z(F) = (Z(F) \cap U_0) \cup (Z(F) \cap H_\infty),$$

so  $Z(F)$  can be larger than the affine set  $Z(f)$  because it may contain additional points at infinity.

### What points at infinity are added

Assume  $\deg(f) \leq 3$  and write  $f$  as a sum of homogeneous parts:

$$f = f_3 + f_2 + f_1 + f_0,$$

where  $f_d$  is homogeneous of degree  $d$ .

The homogenization of degree 3 is

$$F(x_0, x_1, \dots, x_n) = f_3(x_1, \dots, x_n) + x_0 f_2(x_1, \dots, x_n) + x_0^2 f_1(x_1, \dots, x_n) + x_0^3 f_0.$$

Restricting to the hyperplane at infinity  $H_\infty = \{x_0 = 0\}$  gives

$$F(0, x_1, \dots, x_n) = f_3(x_1, \dots, x_n).$$

Therefore the points at infinity of the projective closure are exactly the projective zeros of the top-degree part:

$$Z(F) \cap H_\infty = Z(f_3) \subset \mathbb{P}^{n-1}.$$

So the geometry at infinity is controlled by the leading homogeneous term of  $f$ .

**Example**

Consider the affine cubic curve in  $\mathbb{A}^2$ :

$$f(x, y) = y - x^3.$$

Its top-degree part is

$$f_3(x, y) = -x^3.$$

Homogenizing to degree 3 in  $\mathbb{P}^2$  yields

$$F(x, y, z) = yz^2 - x^3.$$

On the affine chart  $z = 1$ ,

$$F(x, y, 1) = y - x^3,$$

so  $Z(F) \cap \{z \neq 0\}$  is the original affine curve.

At infinity  $z = 0$ ,

$$F(x, y, 0) = -x^3,$$

so

$$x = 0,$$

and the projective closure contains the point

$$(0 : 1 : 0).$$

Thus the projective curve is larger than the affine curve by the additional point at infinity.

**Summary**

Homogeneous polynomials are more restrictive as equations, but projective space contains additional points (the hyperplane at infinity). When one passes from an affine hypersurface  $Z(f) \subset \mathbb{A}^n$  to its projective closure  $Z(F) \subset \mathbb{P}^n$ , one typically adds solutions at infinity, which is why  $Z(F)$  can be larger than  $Z(f)$ .

## Picard group and blow-ups (plane case, with applications to Cremona)

**Divisors and the Picard group**

Let  $X$  be a smooth irreducible variety over an algebraically closed field  $k$ .

**Weil divisors**

A *Weil divisor* on  $X$  is a finite formal integer linear combination of irreducible codimension-1 subvarieties:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The group of Weil divisors is denoted

$$\text{Div}(X).$$

A divisor is called *effective* if

$$n_i \geq 0$$

for all  $i$ , and we write

$$D \geq 0.$$



## Principal divisors

If  $f \in k(X)^\times$  is a nonzero rational function on  $X$ , then its divisor of zeros and poles is a divisor

$$\operatorname{div}(f) \in \operatorname{Div}(X),$$

called a *principal divisor*. Two divisors are *linearly equivalent* if their difference is principal:

$$D \sim D' \iff D - D' = \operatorname{div}(f) \text{ for some } f \in k(X)^\times.$$

## Picard group

The *Picard group* of  $X$  is

$$\operatorname{Pic}(X) = \{\text{line bundles on } X\} / \cong$$

with group law given by tensor product.

On a smooth variety, one has the fundamental identification

$$\operatorname{Pic}(X) \cong \operatorname{Div}(X) / \sim,$$

so the Picard group may be viewed as divisors modulo linear equivalence.

## Picard group of projective space

For projective space,

$$\operatorname{Pic}(\mathbb{P}^n) \cong \mathbb{Z}.$$

A generator is the class of a hyperplane divisor  $H$ . Thus every line bundle is of the form

$$\mathcal{O}_{\mathbb{P}^n}(d), \quad d \in \mathbb{Z},$$

and divisor classes are multiples of  $H$ :

$$\operatorname{Pic}(\mathbb{P}^n) = \mathbb{Z} \cdot H.$$

In particular, on  $\mathbb{P}^2$  we write  $H$  for the class of a line.

## Blow-up at a point: geometric idea

Let  $X$  be a smooth surface and let  $p \in X$  be a point. The *blow-up of  $X$  at  $p$*  is a morphism

$$\pi : \tilde{X} \rightarrow X$$

such that:

1.  $\pi$  is an isomorphism over  $X \setminus \{p\}$ ,
2. the fiber  $\pi^{-1}(p)$  is a copy of  $\mathbb{P}^1$ , called the *exceptional divisor*.

Intuitively, blowing up replaces the point  $p$  by the space of directions through  $p$ . On a surface, directions are parametrized by  $\mathbb{P}^1$ , which explains why the exceptional fiber is  $\mathbb{P}^1$ .

**Local model in affine coordinates**

In local affine coordinates  $(u, v)$  near  $p = (0, 0) \in \mathbb{A}^2$ , the blow-up

$$\mathrm{Bl}_0(\mathbb{A}^2) \subset \mathbb{A}^2 \times \mathbb{P}^1$$

is given by the incidence relation

$$ub = va$$

in coordinates  $(u, v) \in \mathbb{A}^2$  and  $(a : b) \in \mathbb{P}^1$ . The projection

$$\pi(u, v, (a : b)) = (u, v)$$

is the blow-down map. The exceptional divisor is

$$E = \pi^{-1}(0, 0) \cong \mathbb{P}^1.$$

**Picard group under blow-up of  $\mathbb{P}^2$** 

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of  $\mathbb{P}^2$  at one point  $p$ . Let  $E$  be the exceptional divisor.

**Generators and structure**

The Picard group becomes rank 2:

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

where  $H$  denotes the pullback of the class of a line:

$$H = \pi^*(\text{line class}).$$

If we blow up  $r$  distinct points  $p_1, \dots, p_r$ , then

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E_1 \oplus \dots \oplus \mathbb{Z} \cdot E_r.$$

**Intersection numbers**

On a smooth surface, divisor classes have an intersection product. For the blow-up of  $\mathbb{P}^2$  at  $r$  points one has

$$H^2 = 1, \quad H \cdot E_i = 0, \quad E_i \cdot E_j = 0 \text{ for } i \neq j, \quad E_i^2 = -1.$$

The key geometric meaning is:

- $H^2 = 1$  because two general lines meet in one point.
- $E_i^2 = -1$  reflects that  $E_i$  is an exceptional curve of the first kind.

## Strict transforms and multiplicities

Let  $C \subset \mathbb{P}^2$  be a plane curve of degree  $d$ . Assume  $C$  passes through  $p_i$  with multiplicity  $m_i$ . After blowing up at the points  $p_i$ , the class of the strict transform  $\tilde{C}$  in the Picard group is

$$[\tilde{C}] = dH - \sum_{i=1}^r m_i E_i.$$

This formula records two pieces of data:

1. the degree  $d$  of the original curve,
2. how much of  $C$  is concentrated at the blown-up points, via the multiplicities  $m_i$ .

## Cremona involution via blow-ups (why Picard matters)

Let

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$$

be the standard Cremona transformation

$$(x : y : z) \mapsto (yz : zx : xy),$$

with base points

$$p_1 = (1 : 0 : 0), \quad p_2 = (0 : 1 : 0), \quad p_3 = (0 : 0 : 1).$$

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up at  $p_1, p_2, p_3$ , with exceptional divisors  $E_1, E_2, E_3$  and line class  $H$ .

Then  $\sigma$  lifts to a morphism on  $\widetilde{\mathbb{P}^2}$ , and its action on divisor classes is encoded by

$$\sigma^*(H) = 2H - E_1 - E_2 - E_3,$$

$$\sigma^*(E_1) = H - E_2 - E_3, \quad \sigma^*(E_2) = H - E_1 - E_3, \quad \sigma^*(E_3) = H - E_1 - E_2.$$

## Degree change formula

Let  $C \subset \mathbb{P}^2$  be a curve of degree  $d$ , with multiplicities  $m_1, m_2, m_3$  at the base points. The strict transform of  $\sigma(C)$  has degree

$$d' = 2d - m_1 - m_2 - m_3.$$

This is obtained by tracking the class

$$[\tilde{C}] = dH - m_1 E_1 - m_2 E_2 - m_3 E_3$$

under the pullback action on  $\text{Pic}(\widetilde{\mathbb{P}^2})$ .

**Example**

A general line has  $d = 1$  and  $m_1 = m_2 = m_3 = 0$ , hence

$$d' = 2,$$

so it maps to a conic. A conic through all three base points has  $d = 2$  and  $m_1 = m_2 = m_3 = 1$ , hence

$$d' = 1,$$

so it maps to a line.

**Picard group and blow-ups (plane case)****Divisors and the Picard group**

Let  $X$  be a smooth irreducible variety over an algebraically closed field  $k$ .

**Weil divisors**

A *Weil divisor* on  $X$  is a finite formal integer linear combination of irreducible codimension-1 subvarieties:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The group of Weil divisors is denoted

$$\text{Div}(X).$$

**Principal divisors and linear equivalence**

If  $f \in k(X)^\times$  is a nonzero rational function on  $X$ , then  $f$  defines a divisor of zeros and poles

$$\text{div}(f) \in \text{Div}(X),$$

called a *principal divisor*. Two divisors are *linearly equivalent* if their difference is principal:

$$D \sim D' \iff D - D' = \text{div}(f) \text{ for some } f \in k(X)^\times.$$

**Picard group**

The *Picard group* of  $X$  is the group of line bundles on  $X$  up to isomorphism:

$$\text{Pic}(X) = \{\text{line bundles on } X\} / \cong,$$

with group law given by tensor product. On a smooth variety there is an identification

$$\text{Pic}(X) \cong \text{Div}(X) / \sim,$$

so  $\text{Pic}(X)$  can be viewed as divisor classes modulo linear equivalence.

## Picard group of $\mathbb{P}^2$

For projective space one has

$$\mathrm{Pic}(\mathbb{P}^n) \cong \mathbb{Z}.$$

A generator is the hyperplane class  $H$ . In particular,

$$\mathrm{Pic}(\mathbb{P}^2) = \mathbb{Z} \cdot H,$$

and every divisor class on  $\mathbb{P}^2$  is of the form  $dH$ .

### 0.1 Blow-up of a point on a surface

Let  $X$  be a smooth surface and let  $p \in X$ . The *blow-up of  $X$  at  $p$*  is a morphism

$$\pi : \tilde{X} \rightarrow X$$

such that:

1.  $\pi$  is an isomorphism over  $X \setminus \{p\}$ ,
2. the fiber  $\pi^{-1}(p)$  is a curve

$$E = \pi^{-1}(p) \cong \mathbb{P}^1,$$

called the *exceptional divisor*.

Geometrically, blowing up replaces the point  $p$  by the set of tangent directions through  $p$ .

#### 0.1.1 Local model in affine coordinates

In local affine coordinates  $(u, v)$  near  $p = (0, 0) \in \mathbb{A}^2$ , the blow-up

$$\mathrm{Bl}_0(\mathbb{A}^2) \subset \mathbb{A}^2 \times \mathbb{P}^1$$

is given by

$$ub = va$$

in coordinates  $(u, v) \in \mathbb{A}^2$  and  $(a : b) \in \mathbb{P}^1$ . The projection

$$\pi(u, v, (a : b)) = (u, v)$$

is the blow-down map, and the exceptional divisor is  $E = \{u = v = 0\} \cong \mathbb{P}^1$ .

## Picard group after blowing up $\mathbb{P}^2$

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of  $\mathbb{P}^2$  at  $r$  distinct points  $p_1, \dots, p_r$ , with exceptional divisors  $E_1, \dots, E_r$ . Let  $H$  denote the pullback of the class of a line.

Then

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E_1 \oplus \dots \oplus \mathbb{Z} \cdot E_r.$$

**Intersection numbers**

On the blow-up of  $\mathbb{P}^2$  at  $r$  points, the intersection product satisfies

$$H^2 = 1, \quad H \cdot E_i = 0, \quad E_i \cdot E_j = 0 \text{ for } i \neq j, \quad E_i^2 = -1.$$

**Strict transforms and multiplicities**

Let  $C \subset \mathbb{P}^2$  be a plane curve of degree  $d$ . Assume  $C$  passes through  $p_i$  with multiplicity  $m_i$ . Then the divisor class of the strict transform  $\tilde{C} \subset \widetilde{\mathbb{P}^2}$  is

$$[\tilde{C}] = dH - \sum_{i=1}^r m_i E_i.$$

**Remark: connection to Cremona**

For the standard Cremona involution with base points  $p_1, p_2, p_3$ , one blows up these three points to obtain

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2.$$

The Cremona map lifts to a morphism on  $\widetilde{\mathbb{P}^2}$  and acts linearly on

$$\text{Pic}(\widetilde{\mathbb{P}^2}).$$

Tracking a curve class

$$dH - m_1 E_1 - m_2 E_2 - m_3 E_3$$

under this action yields the degree change formula

$$d' = 2d - m_1 - m_2 - m_3.$$

## Charts for blow-ups, exceptional divisor, and Cremona

### Chart 1: Blow-up of a point on a surface

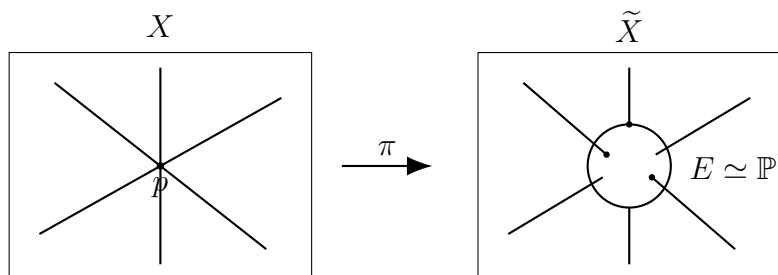


Figure 4: Blowing up replaces the point  $p$  by the exceptional divisor  $E \simeq \mathbb{P}^1$ . Different directions through  $p$  become different points on  $E$ .

### Chart 2: Picard group generators and intersection form on $\text{Bl}_p(\mathbb{P}^2)$

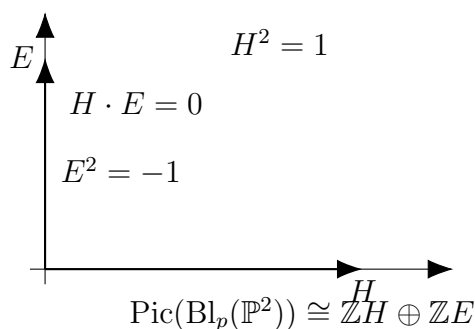


Figure 5: After blowing up one point, the Picard group gains the new generator  $E$  (the exceptional divisor), with intersection numbers  $H^2 = 1$ ,  $H \cdot E = 0$ , and  $E^2 = -1$ .

### Chart 3: Cremona involution and the three base points

## Concrete divisors, codimension, and line bundles

### Why divisors use codimension one

Let  $X$  be an irreducible variety. A *prime divisor* on  $X$  means an irreducible closed subvariety of codimension 1. A *Weil divisor* is a finite formal integer combination of prime divisors:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The reason codimension 1 appears is that zeros and poles of a rational function occur along hypersurfaces. For a nonzero rational function  $f \in k(X)^\times$  one defines the *principal divisor*

$$\text{div}(f) = \sum_V \text{ord}_V(f) V,$$

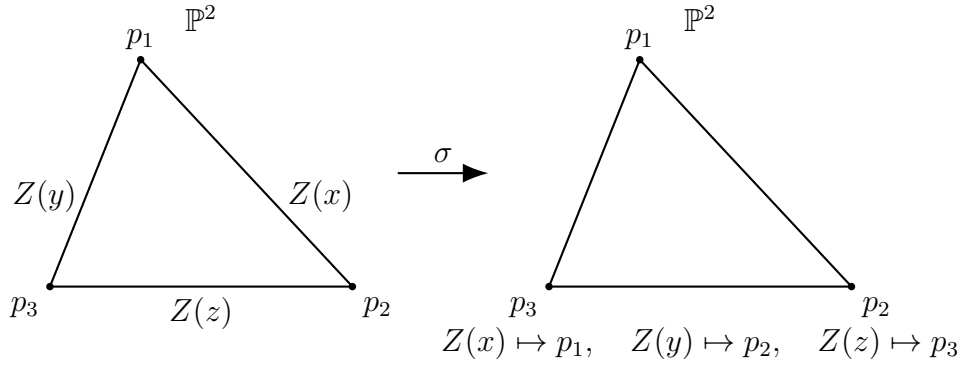


Figure 6: The standard Cremona map has base points  $p_1, p_2, p_3$  and contracts the coordinate lines to these points. After blowing up the base points, the map becomes regular on the blow-up.

where the sum runs over prime divisors  $V \subset X$  and  $\text{ord}_V(f)$  is the order of vanishing (negative for poles) along  $V$ . Higher codimension sets do not contribute to  $\text{div}(f)$ .

In particular:

$$\dim(X) = 1 \implies \text{prime divisors are points,}$$

$$\dim(X) = 2 \implies \text{prime divisors are curves.}$$

### Example: divisors on $\mathbb{P}^1$

On  $X = \mathbb{P}^1$ , prime divisors are points. Using the affine coordinate  $t = x/y$ , consider

$$f(t) = \frac{t-a}{t-b} \in k(t)^\times.$$

Then  $f$  has a simple zero at  $t = a$  and a simple pole at  $t = b$ , hence

$$\text{div}(f) = (a) - (b).$$

One has

$$\text{Pic}(\mathbb{P}^1) \cong \mathbb{Z},$$

and the degree of a divisor

$$D = \sum_i n_i(p_i)$$

is

$$\deg(D) = \sum_i n_i.$$

Every principal divisor has degree 0.

### Example: divisors on $\mathbb{P}^2$

On  $X = \mathbb{P}^2$ , prime divisors are irreducible plane curves. Let  $(x : y : z)$  be homogeneous coordinates and consider the rational function

$$f = \frac{x}{z} \in k(\mathbb{P}^2)^\times.$$



The zero locus of  $f$  is the line  $Z(x)$  and the pole locus is the line  $Z(z)$ , so

$$\operatorname{div}\left(\frac{x}{z}\right) = Z(x) - Z(z).$$

Thus  $Z(x) \sim Z(z)$ , and in fact all lines are linearly equivalent. One has

$$\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z},$$

generated by the class  $H$  of a line.

### Example: blow-up of $\mathbb{P}^2$ at a point

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of  $\mathbb{P}^2$  at a point  $p$ . The exceptional divisor is

$$E = \pi^{-1}(p) \cong \mathbb{P}^1.$$

The Picard group becomes

$$\operatorname{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

where  $H$  is the pullback of the class of a line.

If  $C \subset \mathbb{P}^2$  is a curve of degree  $d$  and has multiplicity  $m$  at  $p$ , then the class of its strict transform satisfies

$$[\widetilde{C}] = dH - mE.$$

For example, a line not passing through  $p$  has strict transform class  $H$ , while a line passing through  $p$  has strict transform class  $H - E$ .

### Line bundles are not projective lines

A *line bundle* on  $X$  is a rank-1 vector bundle. Equivalently, it is a morphism

$$\pi : L \rightarrow X$$

such that for each  $x \in X$  the fiber  $L_x$  is a 1-dimensional  $k$ -vector space and locally

$$L|_U \cong U \times k.$$

Thus a line bundle is a family of 1-dimensional vector spaces parametrized by  $X$ .

The variety  $\mathbb{P}^1$  is a projective line, not a line bundle. A  $\mathbb{P}^1$ -bundle arises as the projectivization  $\mathbb{P}(E)$  of a rank-2 vector bundle  $E$ . If  $L$  is a line bundle (rank 1), then its projectivization has fibers  $\mathbb{P}^0$  (a point), so

$$\mathbb{P}(L) \cong X.$$

### One blow-up of $\mathbb{P}^2$ : concrete divisor classes and computations

Let  $p \in \mathbb{P}^2$  be a point and let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of  $\mathbb{P}^2$  at  $p$ . Let

$$E = \pi^{-1}(p) \cong \mathbb{P}^1$$

be the exceptional divisor. Let  $H$  denote the pullback of the class of a line in  $\mathbb{P}^2$ .

**Picard group and intersection pairing**

One has

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

and the intersection numbers satisfy

$$H^2 = 1, \quad H \cdot E = 0, \quad E^2 = -1.$$

For divisor classes

$$D = aH - bE, \quad D' = cH - dE,$$

the intersection product is

$$D \cdot D' = ac - bd.$$

**Strict transform class, degree, and multiplicity**

Let  $C \subset \mathbb{P}^2$  be a plane curve of degree  $d$ , and assume  $C$  has multiplicity  $m$  at  $p$ . Then the divisor class of the strict transform  $\widetilde{C} \subset \widetilde{\mathbb{P}^2}$  is

$$[\widetilde{C}] = dH - mE.$$

Moreover, degree and multiplicity can be recovered as intersection numbers:

$$\widetilde{C} \cdot H = d, \quad \widetilde{C} \cdot E = m.$$

Indeed,

$$(dH - mE) \cdot H = dH^2 - m(E \cdot H) = d,$$

and

$$(dH - mE) \cdot E = d(H \cdot E) - mE^2 = m.$$

**Examples**

1. A line  $L$  not passing through  $p$  has strict transform class

$$[\widetilde{L}] = H,$$

and

$$\widetilde{L}^2 = H^2 = 1.$$

2. A line  $L$  passing through  $p$  has multiplicity  $m = 1$  at  $p$ , hence

$$[\widetilde{L}] = H - E.$$

Its self-intersection is

$$(H - E)^2 = H^2 - 2(H \cdot E) + E^2 = 1 - 1 = 0,$$

and it meets the exceptional divisor once:

$$(H - E) \cdot E = 1.$$

3. A conic  $Q$  not passing through  $p$  has class

$$[\tilde{Q}] = 2H,$$

so

$$\tilde{Q}^2 = (2H)^2 = 4.$$

4. A conic  $Q$  passing through  $p$  simply has multiplicity  $m = 1$ , hence

$$[\tilde{Q}] = 2H - E.$$

Then

$$(2H - E)^2 = 4 - 1 = 3, \quad (2H - E) \cdot E = 1.$$

5. A conic  $Q$  with multiplicity  $m = 2$  at  $p$  has class

$$[\tilde{Q}] = 2H - 2E,$$

so

$$(2H - 2E)^2 = 4 - 4 = 0, \quad (2H - 2E) \cdot E = 2.$$

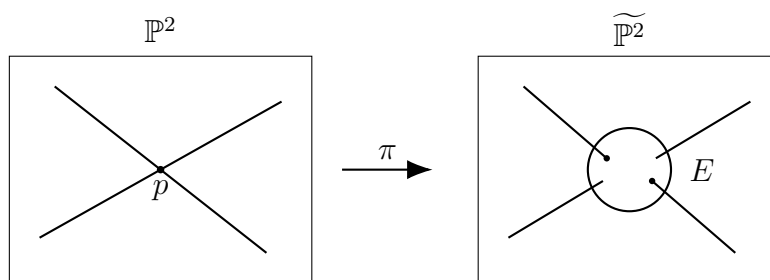


Figure 7: Blowing up replaces  $p$  by  $E \cong \mathbb{P}^1$ ; different directions through  $p$  become different points on  $E$ .

## The tautological line bundle on projective space

Let  $V$  be a finite-dimensional  $k$ -vector space,  $\dim(V) = n + 1$ . Projective space  $\mathbb{P}(V)$  is the set of 1-dimensional linear subspaces  $\ell \subset V$ . A point of  $\mathbb{P}(V)$  is written  $[\ell]$ .

### The tautological bundle $\mathcal{O}_{\mathbb{P}(V)}(-1)$

There is a canonical rank-1 subbundle of the trivial bundle  $\mathbb{P}(V) \times V$ :

$$\mathcal{O}_{\mathbb{P}(V)}(-1) \subset \mathbb{P}(V) \times V,$$

defined by

$$\mathcal{O}_{\mathbb{P}(V)}(-1) = \{ ([\ell], v) \in \mathbb{P}(V) \times V \mid v \in \ell \}.$$

Its fiber over a point  $[\ell] \in \mathbb{P}(V)$  is the line itself:

$$\mathcal{O}_{\mathbb{P}(V)}(-1)_{[\ell]} = \ell.$$

This is why it is called the tautological line bundle.

### The principal $k^\times$ -bundle and the relation to $\mathcal{O}(-1)$

There is a natural quotient map

$$V \setminus \{0\} \longrightarrow \mathbb{P}(V), \quad v \longmapsto [kv],$$

whose fibers are  $\ell \setminus \{0\}$ . After choosing a basis of  $\ell$ , one identifies  $\ell \setminus \{0\}$  with  $k^\times$ . This makes  $V \setminus \{0\} \rightarrow \mathbb{P}(V)$  a principal  $k^\times$ -bundle. The line bundle  $\mathcal{O}_{\mathbb{P}(V)}(-1)$  can be viewed as the associated line bundle obtained by adjoining the zero vector in each fiber, but the intrinsic definition above is standard.

### The hyperplane bundle $\mathcal{O}(1)$ and the Picard group

Define the dual line bundle

$$\mathcal{O}_{\mathbb{P}(V)}(1) = \mathcal{O}_{\mathbb{P}(V)}(-1)^\vee.$$

On  $\mathbb{P}^n = \mathbb{P}(k^{n+1})$ , one has

$$\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z},$$

and the class of  $\mathcal{O}(1)$  is a generator. Hyperplanes are precisely the zero loci of global sections of  $\mathcal{O}(1)$ , and more generally

$$\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$$

corresponds to degree- $d$  hypersurfaces.

### Local trivializations on standard charts

Let  $[x_0 : \cdots : x_n]$  be homogeneous coordinates on  $\mathbb{P}^n$  and set

$$U_i = \{x_i \neq 0\}.$$

On each  $U_i$ , the bundle  $\mathcal{O}(-1)$  is trivial. On overlaps  $U_i \cap U_j$ , the transition function for  $\mathcal{O}(-1)$  is

$$g_{ij} = \frac{x_j}{x_i}.$$

For the dual bundle  $\mathcal{O}(1)$  the transition function is

$$g_{ij}^{-1} = \frac{x_i}{x_j}.$$

### Connection to blow-ups

If

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-up at a point, then the line class  $H$  used in Picard computations on  $\widetilde{\mathbb{P}^2}$  is the pullback of the hyperplane bundle:

$$H = \pi^* \mathcal{O}_{\mathbb{P}^2}(1).$$

### From divisors to line bundles: $\mathcal{O}(D)$ and zero loci of sections

Let  $X$  be a smooth irreducible variety over an algebraically closed field  $k$ . On a smooth variety, divisor classes and line bundles are closely linked.

### Cartier divisors and local equations

A Cartier divisor  $D$  on  $X$  can be described by an open cover  $\{U_i\}$  and rational functions

$$f_i \in k(X)^\times$$

such that on overlaps  $U_i \cap U_j$  one has

$$f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Intuitively, each  $f_i$  is a local defining equation for the divisor on  $U_i$ , and the compatibility says that two local equations differ by a nowhere-vanishing regular function.

On a smooth variety every Weil divisor is Cartier, so the construction below applies to Weil divisors as well.

### Definition of the line bundle $\mathcal{O}(D)$

Given a Cartier divisor  $D$  represented by local equations  $\{(U_i, f_i)\}$ , define a line bundle  $\mathcal{O}(D)$  by gluing trivial bundles on each  $U_i$  with transition functions

$$g_{ij} = \frac{f_i}{f_j} \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Equivalently, one may define the associated invertible sheaf by

$$\mathcal{O}(D)(U) = \{h \in k(X) \mid hf_i \in \mathcal{O}_X(U \cap U_i) \text{ for all } i\}.$$

This construction depends only on the linear equivalence class of  $D$ .

### Linear equivalence corresponds to isomorphic line bundles

If

$$D \sim D',$$

then

$$D - D' = \text{div}(\varphi)$$

for some  $\varphi \in k(X)^\times$ . In this case one has a canonical isomorphism of line bundles

$$\mathcal{O}(D) \cong \mathcal{O}(D').$$

Hence the assignment  $D \mapsto \mathcal{O}(D)$  factors through divisor classes and gives the identification

$$\text{Pic}(X) \cong \text{Div}(X) / \sim.$$

### Sections and divisors of zeros and poles

A rational section  $s$  of  $\mathcal{O}(D)$  can be viewed locally as a rational function multiplied by the local trivializing section. Its divisor of zeros and poles, denoted  $\text{div}(s)$ , is a Weil divisor. There is a fundamental relation:

$$\text{div}(s) + D \geq 0 \iff s \text{ is a regular section of } \mathcal{O}(D).$$

In particular, if  $s$  is a nonzero regular section of  $\mathcal{O}(D)$ , then its zero locus defines an effective divisor and one has

$$Z(s) = \text{div}(s) + D.$$

Thus divisors can be realized as zero loci of sections of associated line bundles.

**Example:  $\mathbb{P}^n$ , hyperplanes, and  $\mathcal{O}(1)$** 

Let  $H \subset \mathbb{P}^n$  be a hyperplane. Then the associated line bundle is the hyperplane bundle:

$$\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1).$$

A hyperplane  $H = Z(L)$  is given by a nonzero linear form  $L$ , and  $L$  defines a global section of  $\mathcal{O}(1)$  whose zero locus is exactly  $H$ :

$$Z(L) = H.$$

More generally, for  $d \geq 1$ ,

$$\mathcal{O}_{\mathbb{P}^n}(d) = \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d},$$

and homogeneous polynomials of degree  $d$  define global sections of  $\mathcal{O}(d)$  whose zero loci are degree- $d$  hypersurfaces.

**Connection to blow-ups**

If

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-up at a point, then the class  $H$  used in divisor computations on  $\widetilde{\mathbb{P}^2}$  is the pullback of the hyperplane bundle:

$$\mathcal{O}_{\widetilde{\mathbb{P}^2}}(H) \cong \pi^* \mathcal{O}_{\mathbb{P}^2}(1).$$

**Projective bundles  $\mathbb{P}(E) \rightarrow X$** 

Let  $X$  be a variety and let

$$p : E \rightarrow X$$

be a vector bundle (locally free sheaf) of rank  $r$ . For each point  $x \in X$ , the fiber  $E_x$  is an  $r$ -dimensional  $k$ -vector space.

**Definition of the projective bundle**

The *projective bundle* associated to  $E$  is a morphism

$$\pi : \mathbb{P}(E) \rightarrow X$$

whose fiber over  $x$  is the projective space of the fiber:

$$\pi^{-1}(x) \cong \mathbb{P}(E_x) \cong \mathbb{P}^{r-1}.$$

Thus:

$$\begin{aligned} \text{rank}(E) = 2 &\implies \mathbb{P}(E) \rightarrow X \text{ is a } \mathbb{P}^1\text{-bundle,} \\ \text{rank}(E) = 1 &\implies \mathbb{P}(E) \rightarrow X \text{ has fibers } \mathbb{P}^0 \text{ and } \mathbb{P}(E) \cong X. \end{aligned}$$

**Local triviality**

Projective bundles are locally trivial in the Zariski topology. If  $E$  is trivial on an open set  $U \subset X$ , meaning

$$E|_U \cong U \times k^r,$$

then

$$\mathbb{P}(E)|_U \cong U \times \mathbb{P}^{r-1}.$$

### Tautological line bundle and the universal exact sequence

There are two common conventions:

- $\mathbb{P}(E)$  parameterizes lines in the fibers  $E_x$ ,
- $\mathbb{P}(E)$  parameterizes 1-dimensional quotients of the fibers  $E_x$ .

Using the quotient convention, there is a canonical surjection on  $\mathbb{P}(E)$  producing the *universal quotient line bundle*

$$\pi^*E \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1),$$

and a short exact sequence

$$0 \rightarrow \mathcal{S} \rightarrow \pi^*E \rightarrow \mathcal{O}_{\mathbb{P}(E)}(1) \rightarrow 0,$$

where  $\mathcal{S}$  is a rank  $r - 1$  subbundle. The dual line bundle

$$\mathcal{O}_{\mathbb{P}(E)}(-1) = \mathcal{O}_{\mathbb{P}(E)}(1)^\vee$$

is often called the tautological line bundle.

### Picard group of a projective bundle

Under mild hypotheses (for example  $X$  integral and  $E$  locally free), one has an isomorphism

$$\mathrm{Pic}(\mathbb{P}(E)) \cong \mathrm{Pic}(X) \oplus \mathbb{Z} \cdot [\mathcal{O}_{\mathbb{P}(E)}(1)].$$

Concretely, every line bundle on  $\mathbb{P}(E)$  can be written uniquely as

$$\pi^*L \otimes \mathcal{O}_{\mathbb{P}(E)}(m), \quad L \in \mathrm{Pic}(X), \quad m \in \mathbb{Z}.$$

### Example: blow-up exceptional divisor

If

$$\pi : \tilde{X} \rightarrow X$$

is the blow-up of a smooth variety  $X$  at a point  $p$ , then the exceptional divisor is a projective space:

$$E = \pi^{-1}(p) \cong \mathbb{P}^{\dim(X)-1}.$$

In particular, for the blow-up of  $\mathbb{P}^2$  at a point one has

$$E \cong \mathbb{P}^1.$$

Geometrically,  $E$  parameterizes tangent directions at the blown-up point.

### Concrete examples: sections of $\mathcal{O}(1)$ and $\mathcal{O}(d)$

Let  $[x_0 : \cdots : x_n]$  be homogeneous coordinates on  $\mathbb{P}^n$ .

**Hyperplanes as zero loci of sections of  $\mathcal{O}(1)$** 

Let

$$L = a_0x_0 + \cdots + a_nx_n$$

be a nonzero linear form. The hyperplane is

$$H = Z(L) \subset \mathbb{P}^n.$$

The polynomial  $L$  defines a global section of  $\mathcal{O}_{\mathbb{P}^n}(1)$ , and its vanishing locus as a section is exactly  $H$ :

$$Z(L) = H.$$

On the standard chart

$$U_i = \{x_i \neq 0\},$$

the section is represented by the regular function

$$\frac{L}{x_i}.$$

Hence  $H \cap U_i$  is cut out by one equation

$$\frac{L}{x_i} = 0$$

in the affine coordinates on  $U_i$ .

**Degree- $d$  hypersurfaces as zero loci of sections of  $\mathcal{O}(d)$** 

Let

$$F \in k[x_0, \dots, x_n]$$

be homogeneous of degree  $d \geq 1$ . Then  $F$  defines a global section of

$$\mathcal{O}_{\mathbb{P}^n}(d), \quad \mathcal{O}_{\mathbb{P}^n}(d) = \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d},$$

and its zero locus is the degree- $d$  hypersurface

$$Z(F) \subset \mathbb{P}^n.$$

On the chart  $U_i$  the section is represented by the regular function

$$\frac{F}{x_i^d}.$$

**A principal divisor: why hyperplanes are linearly equivalent**

Consider the rational function on  $\mathbb{P}^n$

$$f = \frac{x_0}{x_1}.$$

Its divisor of zeros and poles is

$$\operatorname{div}(f) = Z(x_0) - Z(x_1).$$

Hence

$$Z(x_0) \sim Z(x_1),$$

so all hyperplanes have the same divisor class in  $\operatorname{Pic}(\mathbb{P}^n)$ , corresponding to  $\mathcal{O}(1)$ .



## Concrete projective bundle example via polynomials: blow-up of $\mathbb{P}^2$ at a point

Let

$$p = (0 : 0 : 1) \in \mathbb{P}^2.$$

The blow-up of  $\mathbb{P}^2$  at  $p$  can be realized as a closed subvariety of  $\mathbb{P}^2 \times \mathbb{P}^1$ :

$$\widetilde{\mathbb{P}^2} = \{((x : y : z), (s : t)) \in \mathbb{P}^2 \times \mathbb{P}^1 \mid xt - ys = 0\}.$$

The projection

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-down map. The exceptional divisor is

$$E = \pi^{-1}(p) = \{((0 : 0 : 1), (s : t))\} \cong \mathbb{P}^1.$$

The second projection

$$\rho : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^1, \quad ((x : y : z), (s : t)) \longmapsto (s : t),$$

exhibits  $\widetilde{\mathbb{P}^2}$  as a  $\mathbb{P}^1$ -bundle over  $\mathbb{P}^1$ . Indeed, for fixed  $(s : t)$  the fiber is

$$\rho^{-1}(s : t) = \{(x : y : z) \in \mathbb{P}^2 \mid tx - sy = 0\},$$

which is a projective line in  $\mathbb{P}^2$ , hence isomorphic to  $\mathbb{P}^1$ .